

Build a DBpedia Version for the Vietnamese Language

Countries from Vietnamese Wikipedia as 5-Star Linked Open Data

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Course

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The DBpedia Project

- A global community effort to extract structured knowledge from Wikipedia.
- The core of the Linked Open Data (LOD) cloud.

Motivation for VN-DBpedia

- While English DBpedia is vast, localized knowledge often lacks structured, semantic representations.
- We need a Vietnamese-language Knowledge Graph to power local semantic search, NLP tasks, and regional Linked Data applications.

Project Objectives (Capstone Requirements)

- ➊ **Define an Ontology:** Represent country data and relationships semantically.
- ➋ **Collect Data:** Scrape infoboxes from Vietnamese Wikipedia.
- ➌ **4-Star Data Standard:** Transform collected JSON data into W3C standard RDF.
- ➍ **Link to English DBpedia:** Map local entities to the global LOD cloud (achieving 5-star standard).
- ➎ **Query Interface:** Host the semantic graph via a SPARQL endpoint.

Our pipeline is divided into two primary phases: Data Engineering and Data Hosting.

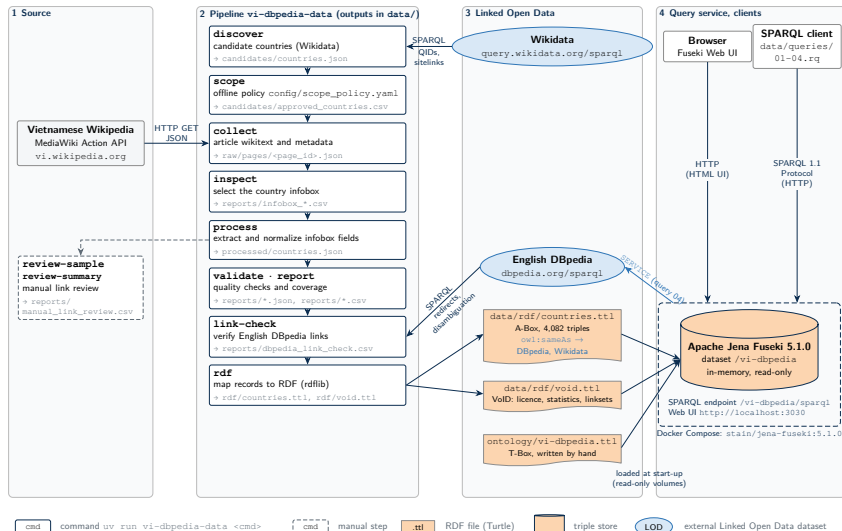
1. Python Extraction Pipeline

- **Discovery:** Wikidata API querying.
- **Extraction:** MediaWiki API (Wikitext to JSON).
- **Transformation:** rdflib generates RDF/Turtle.

2. SPARQL Hosting Service

- **Engine:** Apache Jena Fuseki.
- **Deployment:** Docker Compose.
- **Interface:** Web UI for standard SPARQL querying.

Service Architecture



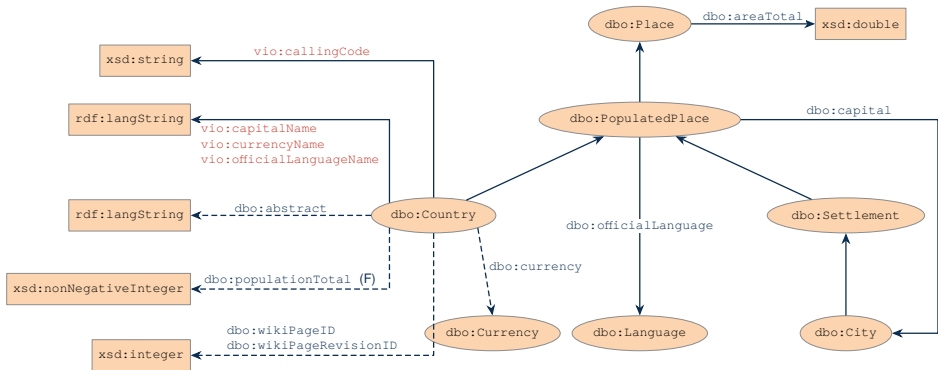
Ontology Design: Reusing DBpedia Standards

Instead of inventing proprietary schemas, we reuse the standard DBpedia (dbo:) ontology and add vio: terms only where DBpedia has none (ontology/vi-dbpedia.ttl).

Raw Field	RDF Predicate	Datatype / Object
Entity URI	vir:Việt_Nam	dbo:Country
Title	rdfs:label	Literal (@vi)
Population	dbo:populationTotal	xsd:nonNegativeInteger
Area (m^2)	dbo:areaTotal	xsd:double
Capital	dbo:capital	Object URI (vir:Hà_Nội, dbo:City)
Calling code	vio:callingCode	xsd:string (new term)

Note: All text literals are strictly tagged with @vi for internationalization.

Ontology (T-Box): ontology/vi-dbpedia.ttl



Class class (owl:Class); range of an owl:ObjectProperty

xsd:... datatype: range of an owl:DatatypeProperty

unlabelled arrow = rdfs:subClassOf

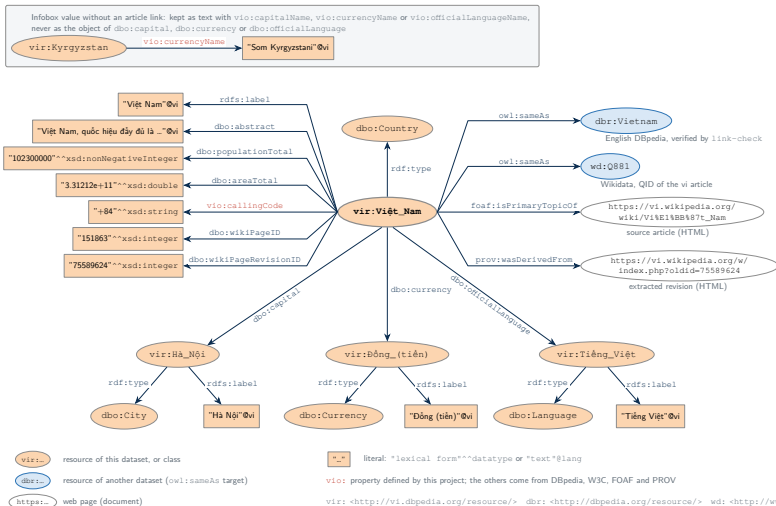
solid arrow property, from its rdfs:domain to its rdfs:range

dashed arrow no rdfs:domain declared; drawn from the class it is used on

dbo: reused DBpedia term **vio:** new term of this project **(F)** owl:FunctionalProperty

dbo: = <http://dbpedia.org/ontology/>, re-declared with the domain, range and hierarchy of the DBpedia Ontology; vio: = <http://vi.dbpedia.org/ontology/>. Also used in the data (W3C, FOAF and PROV vocabularies, not re-declared): rdf:type, rdfs:label, owl:sameAs, foaf:isPrimaryTopicOf, prov:wasDerivedFrom.

Data Schema (A-Box): One Country Record



Whole file `data/rdf/countries.ttl`: 4,082 triples; 197 `dbo:Country`, 199 `dbo:City`, 139 `dbo:Currency`, 169 `dbo:Language`; `owl:sameAs`: 196 to English DBpedia, 197 to Wikidata.
The abstract literal is shortened in this figure.

Achieving 4-Star and 5-Star Linked Data

4-Star Data (RDF Serialization)

Data is published in non-proprietary Turtle (.ttl) format with strict typing.

5-Star Data (Cross-Lingual Linking)

English DBpedia links come from Wikipedia interlanguage links, verified against the DBpedia SPARQL endpoint (redirects resolved, disambiguation pages dropped); every country is also linked to Wikidata.

Turtle (data/rdf/countries.ttl)

```
vir:Việt_Nam a dbo:Country ;  
    rdfs:label "Việt Nam"@vi ;  
    dbo:populationTotal "102300000"^^xsd:nonNegativeInteger ;  
    owl:sameAs dbr:Vietnam , wd:Q881 .
```

Dataset Metrics:

- **197** Sovereign States processed.
- **4,082** RDF triples describing **704** typed entities.
- **196/197** verified links to English DBpedia, **197/197** to Wikidata.

SPARQL Web Service:

- Hosted entirely on a local Dockerized Apache Jena Fuseki server.
- Allows users to run advanced relational, filter-based, and mathematical graph queries instantaneously.

Summary

- Successfully mapped unstructured Vietnamese Wikitext into a formal, 5-Star Semantic Web graph.
- Fulfilled all 5 capstone requirements.

Future Work

- Expand the pipeline domain (e.g., Cities, People, Organizations).
- Implement automated daily syncs with Wikipedia updates.

Thank You

Q & A